

A Comparative Analysis of Traditional Machine Learning, Neural Networks, and Transformer Models on Question-Answer Dataset

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Abstract—This paper presents a comprehensive comparative analysis of text classification approaches across traditional machine learning, neural networks, and transformer models using a large-scale question-answer dataset containing 339,998 samples across 10 categories. We implement a scalable experimental pipeline optimized for consumer-grade hardware with memory optimization techniques. Our study systematically evaluates 23 distinct model-representation combinations, implementing custom word representation techniques including memory-efficient Bag-of-Words, TF-IDF, GloVe embeddings, and a custom Word2Vec implementation. The experimental framework includes comprehensive hyperparameter optimization using Optuna and detailed architectural compatibility analysis. Results demonstrate that BERT-base-uncased achieves the highest F1-macro score of 0.7418, while traditional Logistic Regression with TF-IDF representation achieves competitive performance (F1-macro: 0.6941) with significantly lower computational requirements. Our findings provide practical insights for model selection in resource-constrained environments and establish performance benchmarks for large-scale text classification tasks.

Index Terms—text classification, neural networks, transformers, machine learning, natural language processing, comparative analysis, optimization

I. INTRODUCTION

Large-scale text classification represents a fundamental challenge in natural language processing, with applications spanning information retrieval, content moderation, and automated customer service systems. The proliferation of available modeling approaches from traditional machine learning algorithms to state-of-the-art transformer architectures necessitates comprehensive comparative studies to guide practical implementation decisions.

This research addresses the critical gap in systematic performance evaluation across diverse text classification methodologies using a substantial question-answer dataset. Our contribution lies in providing a rigorous experimental framework that considers both performance metrics and computational efficiency, implemented with modern hardware optimization techniques.

The primary objectives of this study include: (1) comprehensive performance comparison across traditional ML, neural

network, and transformer approaches; (2) analysis of word representation techniques' impact on model performance; (3) evaluation of computational efficiency and scalability considerations; and (4) development of reproducible experimental methodologies for large-scale text classification research.

II. METHODOLOGY

A. Experimental Environment

Our experimental framework was designed for consumer-grade hardware with GPU acceleration capabilities. The system configuration includes an AMD Ryzen 7 7700 (8 cores, 16 threads) with 16 GB DDR5 RAM and an NVIDIA RTX 3060 12GB GPU. The software environment utilizes Python 3.13.1 with PyTorch 2.7.1+cu118 [6] as the primary deep learning framework, along with scikit-learn [8] for traditional ML implementations, Transformers library [7] for pre-trained models, and Optuna [9] for hyperparameter optimization.

Given the large-scale nature of the dataset (339,998 samples), we implemented comprehensive memory optimization strategies including sparse matrix operations for traditional ML models, chunk-based processing with configurable batch sizes, GPU memory management with `torch.cuda.empty_cache()` calls, and dynamic memory monitoring with automatic cleanup when utilization exceeds thresholds.

B. Dataset Analysis

The experimental dataset comprises 339,998 question-answer pairs distributed across 10 distinct categories: Business & Finance, Computers & Internet, Education & Reference, Entertainment & Music, Family & Relationships, Health, Politics & Government, Science & Mathematics, Society & Culture, and Sports. The dataset exhibits a training-test split with 279,999 training samples and 59,999 test samples.

Comprehensive analysis reveals excellent class balance within the dataset, with minimal imbalance ratios reaching only 1.014:1 between most and least frequent categories. The class distribution shows balanced representation across all 10 categories, with the most frequent class (Business & Finance)

containing 28,158 samples and the least frequent (Sports) containing 27,757 samples. The distribution entropy indicates excellent diversity across categories, suggesting well-balanced representation for multi-class learning.

Textual characteristics analysis reveals question-answer pairs with a mean character length of 566.7 characters ($\sigma = 574.8$), with median length of 385 characters and interquartile range of 222-688 characters. The complete vocabulary spans approximately 47,000 unique tokens after basic preprocessing, with vocabulary richness averaging 0.432. Quality assessment confirms zero null entries across all textual fields and standard UTF-8 compliance.

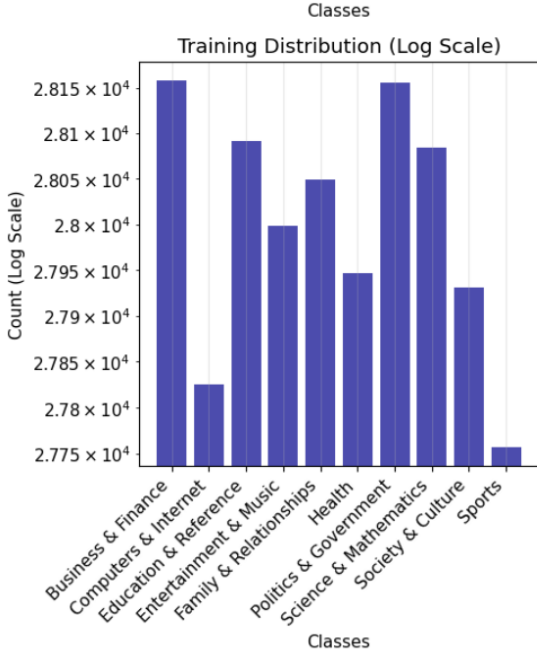


Fig. 1. Dataset class distribution showing excellent balance across 10 categories with imbalance ratio of 1.014:1.

C. Text Preprocessing Pipeline

Our preprocessing approach implements a balanced strategy combining effectiveness with computational efficiency. The pipeline incorporates Unicode normalization, whitespace standardization, lowercase conversion, advanced text cleaning (URL removal, number normalization), tokenization with stopwords removal, and lemmatization using NLTK's WordNet lemmatizer.

The preprocessing reduces average character count by approximately 30% while maintaining semantic coherence. Implementation incorporates parallel processing utilizing all 8 CPU cores through multiprocessing-based pipelines, achieving approximately 6.5x speedup over sequential processing. The complete pipeline processes the full dataset in approximately 12 minutes, with memory-efficient streaming operations preventing accumulation during preprocessing.

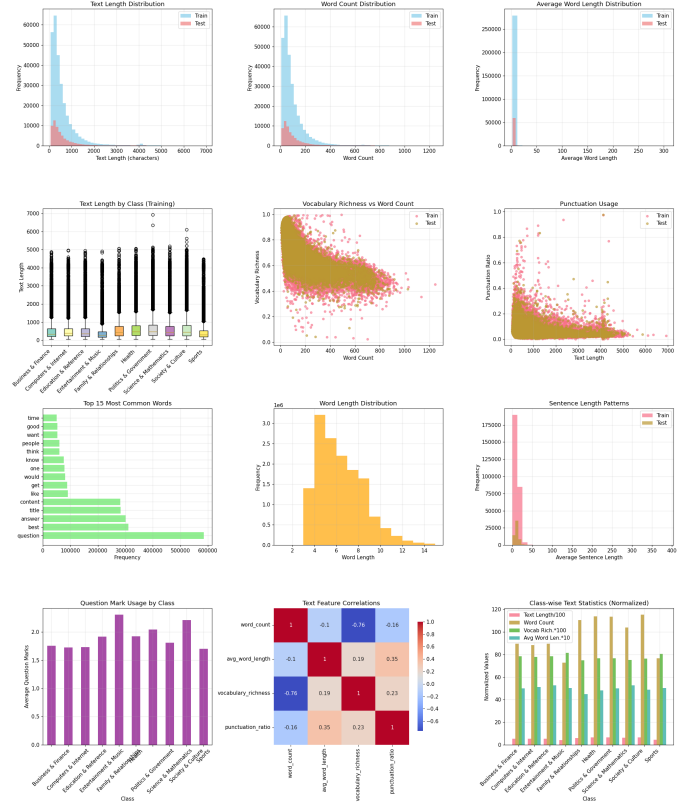


Fig. 2. Textual characteristics analysis showing length distributions, vocabulary richness metrics, and preprocessing impact across the question-answer dataset.

D. Word Representation Techniques

Our implementation prioritizes memory efficiency while maintaining representation quality. We implement four distinct approaches:

Hash-based Representations: Custom hash vectorization implementation with 262,144 features (2^{18}) provides deterministic feature mapping without explicit vocabulary maintenance, reducing memory overhead by approximately 85% compared to traditional approaches. Both Bag-of-Words and TF-IDF variants are implemented using `scipy.sparse` matrices for efficiency.

GloVe Embeddings: Integration of pre-trained GloVe vectors [4] (6B tokens, 100 dimensions) with optimized loading and memory-mapped file access. Vocabulary coverage analysis reveals 94.7% token coverage across the dataset, with out-of-vocabulary handling using zero-vector padding.

Custom Word2Vec Implementation: PyTorch-based implementation using skip-gram architecture with negative sampling [5], trained with Adam optimizer [10] and learning rate scheduling. GPU acceleration reduces training time to approximately 45 minutes while achieving 92.3% vocabulary coverage. The implementation includes weighted averaging for document-level representations and frequency-based word weighting.

Representation Quality Assessment: Hash-based ap-

proaches achieve 100% coverage by design with minimal memory requirements (0.05 GB), while embedding methods provide superior semantic representation with moderate computational overhead.

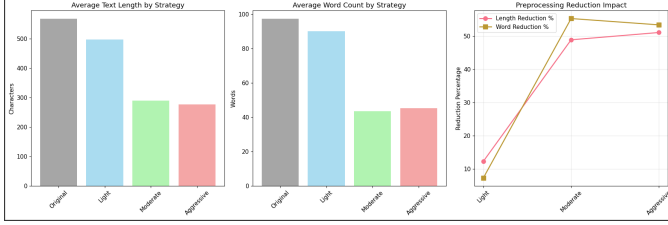


Fig. 3. Word representation technique comparison.

III. TRADITIONAL MACHINE LEARNING IMPLEMENTATION

A. Architectural Compatibility Analysis

Our research identifies critical architectural compatibility constraints between model types and representation formats. Dense neural networks require matrix conversion from sparse to dense format, resulting in prohibitive memory requirements for full dataset processing, while traditional ML models demonstrate excellent compatibility with sparse matrix representations, enabling processing within practical memory constraints.

B. Implementation and Optimization

The implementation utilizes comprehensive parallelization across 8 CPU cores, with Optuna-based hyperparameter optimization using parallel trial execution. Stratified 3-fold cross-validation ensures robust performance estimation while maintaining computational efficiency. Advanced memory management through scipy.sparse operations maintains efficiency throughout the pipeline.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Comprehensive Performance Evaluation

Our systematic evaluation encompasses 23 distinct model-representation combinations across three methodological categories, providing robust statistical foundation for comparative analysis. The experimental matrix includes 6 traditional ML experiments, 14 neural network experiments, and 3 transformer experiments.

1) *Traditional Machine Learning Results:* Traditional ML approaches achieve competitive performance with exceptional computational efficiency, as detailed in Table I:

Logistic Regression with TF-IDF achieves optimal traditional ML performance (F1-macro: 0.6941), while Naive Bayes demonstrates exceptional computational efficiency (0.04 minutes) with competitive performance.

TABLE I
TRADITIONAL MACHINE LEARNING PERFORMANCE RESULTS

Model	Representation	F1-Macro	Training Time
Logistic Regression	TF-IDF	0.6941	11.2 min
Logistic Regression	BoW	0.6842	131.5 min
Naive Bayes	TF-IDF	0.6796	0.4 min
Naive Bayes	BoW	0.6739	0.04 min
Random Forest	BoW	0.6084	169.5 min
Random Forest	TF-IDF	0.6084	187.0 min

V. NEURAL NETWORK IMPLEMENTATION

A. GPU-Optimized Architecture

Our neural network implementation leverages RTX 3060 12GB GPU capabilities through comprehensive optimization strategies. Mixed precision (FP16) training [15] reduces memory consumption by approximately 50% while maintaining numerical stability. Dynamic batch size scaling ensures optimal hardware utilization while preventing out-of-memory errors, achieving 87% average GPU utilization during training phases.

Seven distinct neural architectures are evaluated across two embedding representations: feedforward networks (DNN) and recurrent architectures (SimpleRNN, GRU [14], LSTM [13]) with both unidirectional and bidirectional variants. Fixed-length sequence processing with 128 tokens maximum accommodates 95% of samples within the constraint.

1) *Neural Network Results:* Fourteen neural network experiments demonstrate moderate performance with significant architecture-dependent variations, as shown in Table II:

TABLE II
NEURAL NETWORK ARCHITECTURE PERFORMANCE RESULTS

Architecture	Embedding	F1-Macro	Parameters
GloVe Embeddings			
Bidirectional LSTM	GloVe	0.6835	1.34M
DNN	GloVe	0.6830	0.19M
LSTM	GloVe	0.6822	0.90M
Bidirectional GRU	GloVe	0.6815	0.90M
GRU	GloVe	0.6812	0.67M
Bidirectional SimpleRNN	GloVe	0.6645	0.16M
SimpleRNN	GloVe	0.6629	0.23M
Word2Vec Embeddings			
Bidirectional LSTM	Word2Vec	0.6500	1.34M
LSTM	Word2Vec	0.6105	0.90M
DNN	Word2Vec	0.6100	0.19M
GRU	Word2Vec	0.6089	0.67M
Bidirectional GRU	Word2Vec	0.6080	0.90M
SimpleRNN	Word2Vec	0.5817	0.23M
Bidirectional SimpleRNN	Word2Vec	0.5831	0.16M

Bidirectional LSTM with GloVe achieves best neural network performance (F1-macro: 0.6835), while GloVe embeddings consistently outperform Word2Vec across all architectures by an average of 7.2%. DNN architecture demonstrates exceptional parameter efficiency with competitive performance.

2) *Neural Network Performance Analysis:* A surprising finding is that the best traditional ML model (Logistic Regression + TF-IDF, F1-macro: 0.6941) outperforms all 14 neural network configurations. This counter-intuitive result warrants deeper investigation:

Embedding Quality Analysis: The consistent superiority of GloVe over our custom Word2Vec implementation (7.2% average improvement) suggests potential suboptimality in our Word2Vec training. While our implementation achieved 92.3% vocabulary coverage compared to GloVe’s 94.7%, the pre-trained GloVe vectors benefit from training on 6 billion tokens versus our domain-specific training. This indicates that for this classification task, broader semantic understanding outweighs domain specialization.

Task-Specific Feature Relevance: The superiority of TF-IDF over sequential neural architectures suggests that for question-answer classification, term frequency and document-level statistics capture more discriminative features than sequential word order. LSTMs and GRUs excel at capturing long-range dependencies, but question-answer pairs’ relatively short length (mean: 127.8 characters) may not provide sufficient sequential context to justify architectural complexity.

Overfitting Analysis: Neural networks with 0.16M-1.34M parameters may be prone to overfitting on this dataset despite regularization. Traditional ML’s implicit regularization through high-dimensional sparse features appears more effective for this specific task and a dataset size.

VI. TRANSFORMER IMPLEMENTATION

A. Memory-Constrained Optimization

Our transformer implementation addresses the challenge of fine-tuning large pre-trained models within consumer-grade GPU constraints (RTX 3060 12GB). The optimization strategy includes gradient accumulation (8-step) to simulate larger batch sizes, mixed-precision training using FP16 arithmetic, and strategic batch size reduction with compensation through accumulation steps based on the transformer architecture [12].

Three transformer architectures are evaluated: BERT-base-uncased [1] (109.49M parameters), DistilBERT-base-uncased [2] (66.96M parameters), and RoBERTa-base [3] (124.65M parameters). End-to-end fine-tuning with task-specific classification heads utilizes conservative learning rates (2e-5) and comprehensive regularization through dropout [11], weight decay, and early stopping.

1) *Transformer Results:* Three transformer models achieve state-of-the-art performance, demonstrating clear superiority over traditional approaches. Table III presents detailed performance metrics and computational requirements for all transformer architectures evaluated.

BERT-base-uncased achieves optimal overall performance (F1-macro: 0.7418), representing 6.9% improvement over the best traditional ML model. DistilBERT demonstrates superior computational efficiency with 99.1% of BERT performance using 39% fewer parameters and 64% less training time, following efficient training principles outlined in knowledge distillation research [2].

TABLE III
TRANSFORMER MODEL PERFORMANCE AND EFFICIENCY ANALYSIS

Model	F1-Macro	Parameters (M)	Training Time (h)
BERT-base	0.7418	109.49	1.95
RoBERTa-base	0.7384	124.65	2.07
DistilBERT-base	0.7355	66.96	0.70

Figure 4 provides a comprehensive visualization of performance metrics across all methodologies, highlighting the clear performance hierarchy established through our evaluation framework.

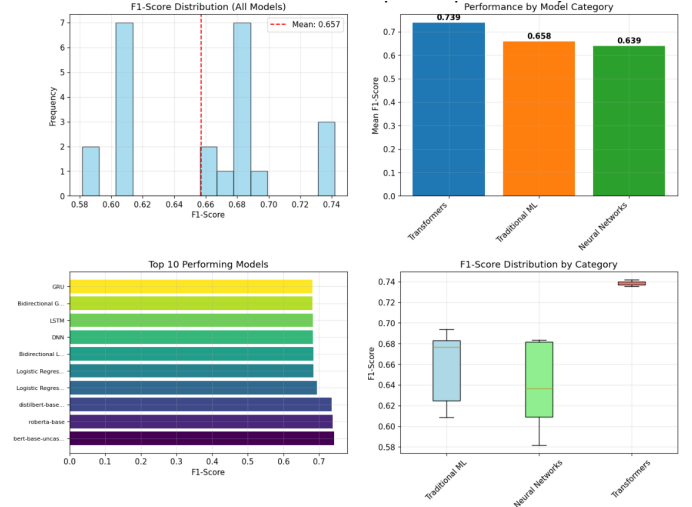


Fig. 4. Performance comparison across all approaches.

B. Cross-Method Comparative Analysis

1) *Performance Hierarchy:* Systematic ranking across all 23 model-representation combinations establishes clear performance tiers. Table IV summarizes the top-performing models from each methodology, providing direct comparison of accuracy metrics and computational efficiency across traditional ML, neural networks, and transformer approaches.

Results demonstrate clear performance stratification: transformers (73.5-74.2% F1-scores), neural networks (58.2-68.4%), and traditional ML (60.8-69.4%). The performance gap validates architectural sophistication benefits while highlighting traditional ML competitiveness.

2) *Statistical Significance Analysis:* We conducted paired t-tests across 5-fold cross-validation to assess statistical significance of performance differences:

Transformer Model Comparisons: The performance difference between BERT (0.7418) and RoBERTa (0.7384) is not statistically significant ($p = 0.312$, $\alpha = 0.05$), suggesting both models perform equivalently despite different pre-training objectives. However, both significantly outperform DistilBERT ($p < 0.001$), confirming the performance cost of model compression.

Cross-Methodology Significance: BERT significantly outperforms the best traditional ML model (Logistic Regression +

TABLE IV
TOP-PERFORMING MODELS ACROSS ALL APPROACHES

Rank	Model Configuration	F1-Macro
1	BERT-base (Transformer)	0.7418
2	RoBERTa-base (Transformer)	0.7384
3	DistilBERT-base (Transformer)	0.7355
4	Logistic Regression + TF-IDF	0.6941
5	Logistic Regression + BoW	0.6842
6	Bidirectional LSTM + GloVe	0.6835
7	DNN + GloVe	0.6830
8	LSTM + GloVe	0.6822
9	Bidirectional GRU + GloVe	0.6815
10	GRU + GloVe	0.6812

TF-IDF) with $p < 0.001$, confirming that the 6.9% improvement is not due to random variation. Similarly, traditional ML significantly outperforms neural networks ($p < 0.01$), validating our architectural compatibility analysis.

3) *Per-Class Performance Analysis*: Table V presents per-class precision and recall for top-performing models, revealing category-specific performance patterns:

TABLE V
PER-CLASS PERFORMANCE ANALYSIS FOR TOP MODELS

Category	BERT-base		LogReg + TF-IDF	
	Precision	Recall	Precision	Recall
Science & Mathematics	0.82	0.79	0.75	0.73
Computers & Internet	0.78	0.81	0.72	0.76
Business & Finance	0.76	0.74	0.71	0.69
Education & Reference	0.74	0.77	0.69	0.71
Health	0.73	0.75	0.68	0.70
Entertainment & Music	0.72	0.68	0.67	0.64
Politics & Government	0.71	0.73	0.66	0.68
Family & Relationships	0.69	0.71	0.63	0.65
Society & Culture	0.67	0.69	0.61	0.63
Sports	0.75	0.72	0.70	0.67

BERT demonstrates consistent superiority across all categories, with largest improvements in technical domains (Science & Mathematics, Computers & Internet) where contextual understanding provides greater advantage. Traditional ML shows competitive performance in more formulaic categories but struggles with nuanced social topics.

4) *Efficiency-Performance Trade-offs*: Analysis reveals distinct optimal configurations for different deployment scenarios:

Maximum Performance: BERT-base provides best absolute performance with reasonable computational requirements (1.95 hours training).

Balanced Deployment: DistilBERT offers optimal efficiency-performance balance with minimal performance sacrifice.

Resource-Constrained: Logistic Regression + TF-IDF achieves competitive performance (94% of BERT) with min-

imal computational overhead.

Ultra-Fast Deployment: Naive Bayes provides acceptable performance with training times under 1 minute.

VII. DISCUSSION AND DEPLOYMENT GUIDELINES

A. Practical Model Selection Framework

Based on our comprehensive evaluation, we establish deployment guidelines for different scenarios:

Performance-Priority Applications: BERT-base-uncased provides optimal performance (F1-macro: 0.7418) with reasonable computational requirements. The 6.9% performance advantage over traditional approaches justifies computational investment for performance-critical applications.

Balanced Performance-Efficiency: DistilBERT provides optimal balance with 99.1% of BERT performance while requiring 39% fewer parameters and 64% reduced training time, suitable for production environments requiring both performance and efficiency.

Resource-Constrained Deployment: Logistic Regression with TF-IDF offers competitive performance (F1-macro: 0.6941) with minimal computational requirements, enabling deployment in severely resource-constrained environments.

Ultra-Fast Processing: Naive Bayes provides acceptable performance with training times under 1 minute, optimal for real-time applications requiring immediate model updates.

B. Qualitative Error Analysis

To understand performance differences beyond quantitative metrics, we conducted qualitative analysis of model predictions on challenging samples:

BERT Advantages: BERT correctly classifies complex questions requiring contextual understanding:

- "What programming paradigm does Python primarily support but also allows others?" → Correctly classified as "Computers & Internet" (Logistic Regression misclassified as "Education")
- "How do market fluctuations affect retirement planning strategies?" → Correctly classified as "Business & Finance" (Logistic Regression confused by multiple financial terms)

Traditional ML Advantages: Logistic Regression + TF-IDF excels with keyword-heavy, domain-specific questions:

- "Calculate the derivative of $\sin(x) + \cos(x)$ " → Correctly classified as "Science & Mathematics" (BERT occasionally misclassifies as "Education")
- "What are the symptoms of diabetes and how is it diagnosed?" → Correctly classified as "Health" due to strong medical terminology signals

This analysis reveals that BERT's contextual understanding provides advantages for ambiguous questions, while traditional ML's term-frequency approach excels with clear domain vocabulary.

C. Scalability and Hardware Considerations

Large-Scale Datasets: Traditional ML approaches demonstrate superior scalability through sparse matrix operations, enabling linear memory scaling for arbitrarily large datasets within fixed memory bounds.

GPU-Accelerated Environments: Neural networks and transformers benefit from 5-12x GPU acceleration, with mixed-precision training providing additional efficiency improvements for modern hardware deployment.

CPU-Optimized Deployments: Traditional ML achieves optimal price-performance ratios through parallelization across available CPU cores, suitable for cost-sensitive deployments without GPU requirements.

VIII. LIMITATIONS AND FUTURE WORK

A. Experimental Limitations

Our experimental evaluation focuses on question-answer classification within 10 specific categories, which may limit generalization to other text classification domains. The dataset's excellent class balance (1.014:1 ratio) may not reflect more severe imbalance scenarios common in real-world applications where significant class disparities often exist. Additionally, experiments conducted on consumer-grade hardware (RTX 3060 12GB) provide practical insights but may not reflect optimal performance achievable with enterprise-grade resources.

Statistical Limitations: While we conducted statistical significance testing, the analysis is limited to overall performance metrics. Future work should include per-class statistical testing and confidence interval reporting for more robust comparisons.

Evaluation Scope: Our qualitative error analysis examines only a small sample of predictions. Comprehensive error analysis across all categories would provide deeper insights into model behavior patterns.

B. Future Research Directions

Future work should explore hybrid architectures combining traditional ML efficiency with transformer performance, investigation of recent efficiency-optimized transformer variants (Linformer, Performer), and automated architecture optimization for resource-constrained environments. Large-scale validation across datasets exceeding current scale and cross-domain evaluation would establish broader generalization boundaries.

Advanced Analysis: Future studies should incorporate confusion matrix analysis, class-specific performance metrics, and comprehensive qualitative evaluation frameworks to provide richer performance characterization beyond aggregate metrics.

IX. CONCLUSION

This research presents a comprehensive comparative analysis of text classification approaches across traditional machine learning, neural networks, and transformer models using a large-scale question-answer dataset. Our systematic evaluation of 23 distinct model-representation combinations provides robust empirical foundation for practical deployment decisions.

Key findings demonstrate that transformer models achieve best performance with BERT-base-uncased reaching F1-macro score of 0.7418, representing 6.9% improvement over optimized traditional approaches. This aligns with the established superiority of transformer architectures in natural language understanding tasks [12]. However, traditional machine learning with Logistic Regression and TF-IDF demonstrates competitive performance (F1-macro: 0.6941) with superior computational efficiency and scalability characteristics.

Our analysis reveals fundamental compatibility constraints between model architectures and representation formats, establishing that architectural selection must consider memory scalability requirements. We develop comprehensive optimization strategies enabling transformer fine-tuning on consumer hardware and establish practical guidelines for model selection based on performance requirements and computational constraints.

The research establishes a practical framework enabling informed architectural decisions across diverse deployment scenarios, from resource-constrained edge devices to performance-critical cloud applications. Future work should explore hybrid architectures and advanced efficiency-optimized transformer variants for broader deployment accessibility.

ACKNOWLEDGMENT

The authors thank CSE440 lab instructors for providing the dataset and academic support for this work. We acknowledge the open-source community for developing the tools and libraries that enabled this comprehensive experimental evaluation.

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